

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic Week 5 Project Summary

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 5 Submission – Data Preparation, Feature Engineering & Baseline Model Development

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1. Introduction

This Week 5 Project Summary accompanies the Data Science track's Data Science Baseline Modelling Report for the HealthConnect Experience Lab. It is deliberately kept concise: it evaluates this week's work against what was planned in the Week 4 Data Science Track Project Plan, and highlights what changed, what was decided, and what remains, without repeating the detailed data-preparation, feature-engineering and evaluation content already documented in the main report.

2. Planned Scope for Week 5

The Week 4 Project Plan's Work Breakdown Structure assigned four work packages to Week 5, under the Analysis and Solution Design phase: resolving open data-quality questions (specifically the status of `waiting_time_minutes`); conducting a formal statistical and correlation analysis; completing feature encoding and scaling; and building and evaluating a baseline logistic regression model. The associated milestone, M2, was defined as “feature set finalised and baseline model trained.”

3. What Was Actually Completed

All four planned work packages were completed. `waiting_time_minutes` was conclusively confirmed as a temporal-leakage risk and excluded from the feature set; categorical features were one-hot encoded and numerical features scaled using training-set-only parameters; and a logistic regression baseline was trained and evaluated on a held-out test set. Milestone M2 is therefore met.

The statistical and correlation analysis work package (2.2) went further than originally scoped: alongside the planned correlation analysis, five independent-samples t-tests were run to formally validate the significance of each candidate numerical feature against the `no_show` target, closing a gap that had been carried as outstanding after the first pass at this work. Three pieces of work went beyond the original Week 5 scope entirely: a patient-aware train/test split was implemented to prevent repeated-patient leakage (see Section 7 of this summary); a previously undocumented data-quality anomaly involving Sunday appointments was identified, investigated and handled; and a documented exchange of findings took place with the Data Analytics track (see Section 7).

4. Key Findings and Development Outcomes

The correlation analysis and the subsequent t-tests confirmed `previous_no_shows` and `booking_lead_days` as the strongest and most statistically significant predictors of no-show behaviour ($p < 0.001$ for both), consistent with the wider no-show prediction literature. `distance_to_clinic_km` and `previous_appointments` were also statistically significant, while `age` sat right at the conventional significance boundary ($p = 0.050$) and is treated as unreliable on its own. The baseline logistic regression model achieved an accuracy of 63.66% and a ROC-AUC of 0.6873 on the held-out test set, meaningfully above random guessing but leaving clear room for improvement. Full metrics, the t-test results and their interpretation are set out in the main report.

5. Major Challenges Encountered

- **Repeated-patient leakage risk:** with 5,000 appointments spread across only 1,696 patients, a standard random train/test split risked placing the same patient's appointments in both sets, which would have overstated the model's real-world performance.
- **An unplanned data-quality anomaly:** 737 appointments were found scheduled on Sundays, despite the Clinic Knowledge Base stating the clinic is closed that day, with no way to confirm from the available material whether this was a data error or an undocumented exception.
- **Internal documentation inconsistency:** an earlier pass at the notebook contained baseline metrics in its narrative commentary and README that did not match the code's actual

executed output. This was substantially corrected during the week: the reported metrics now match the executed results throughout, though the README's description of the train/test split method has not yet been updated to match (see Section 8 of the main report).

6. Important Decisions Made

- **Excluded `waiting_time_minutes` entirely:** rather than imputing it, because it is only known during or after the appointment and would leak future information into a booking-time prediction.
- **Retained Sunday appointments with a flag rather than removing them:** because the discrepancy could not be confirmed as an error against an authoritative source, and removing the records risked discarding genuine information.
- **Selected logistic regression as the baseline model:** for its interpretability, efficiency and native fit for binary classification, to give a transparent benchmark before trying more complex models in Week 6.
- **Treated the executed code output as authoritative:** over the notebook's own stale markdown commentary and README, once the two were found to disagree (see Section 5).

7. Changes to the Week 4 Approach

The Week 4 Machine Learning Problem Definition Document proposed a chronological (time-based) train/test split, so that the model would be evaluated on its ability to predict future appointments from past data. In practice, a *patient-aware grouped split* (grouped by `patient_id`, not by date) was used instead. This was a deliberate change: with 1,696 patients each contributing an average of roughly three appointments, the risk of the same patient appearing in both the training and test sets was judged the more pressing leakage risk to address first. A chronological split remains a valid enhancement and is carried forward as a Week 6 consideration rather than dropped.

8. Cross-Track Collaboration

Unlike the position reported earlier in the week, a documented exchange with the Data Analytics track was completed before this summary was finalised. That track's dashboard-based analysis of the same appointment population independently reported no-show rates by booking lead time, prior no-show history, travel distance and reminder status, using a different tool and framing from the Data Science track's own t-tests and correlation analysis. The two sets of findings corroborate one another, which is a more meaningful form of validation than either track's analysis alone would provide, and is documented in full in Section 8.2 of the main report.

Outputs relevant to the remaining tracks, the confirmed target variable, the final encoded feature set, and the baseline evaluation metrics, remain documented and available to inform the Machine Learning Engineering track's system design and the Project Management track's milestone tracking, consistent with dependencies D1 and D4 in the Week 4 Risk and Dependency Register. The Sunday-appointment anomaly is also relevant to the Generative AI track, since it touches the same Clinic Knowledge Base that track relies on.

9. Remaining Work

A narrower set of items now remains outstanding than at the start of the week: chi-square testing of categorical variables (gender, `appointment_type`, `reminder_sent`), to complete the statistical validation started with this week's t-tests; calibration of the classification threshold against the clinic's actual costs of false positives and negatives; a final correction to the notebook's README so its description of the train/test split matches the patient-aware method actually used; and escalation of the Sunday-appointment anomaly to the data owner or mentor for confirmation.

10. Proposed Focus for Week 6

Week 6 will prioritise training and comparing non-linear candidate models (Random Forest and gradient boosting) against the same patient-aware split, using group-aware cross-validation for any hyperparameter tuning, followed by threshold calibration, chi-square testing of the remaining categorical variables, and the final README correction identified in Section 9.

References

AnalystLab Africa. (2026). *HealthConnect Clinic: Data science track project plan*. Week 4 submission.

AnalystLab Africa. (2026). *HealthConnect Clinic: Data science baseline modelling report*. Week 5 submission.

Dantas, L. F., Fleck, J. L., Cyrino Oliveira, F. L., & Hamacher, S. (2018). *No-shows in appointment scheduling – a systematic literature review*. *Health Policy*, 122(4), 412–421.
<https://doi.org/10.1016/j.healthpol.2018.02.002>

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). *Leakage in data mining: Formulation, detection, and avoidance*. *ACM Transactions on Knowledge Discovery from Data*, 6(4), Article 15. <https://doi.org/10.1145/2382577.2382579>

Saeb, S., Lonini, L., Jayaraman, A., Mohr, D. C., & Kording, K. P. (2017). *The need to approximate the use-case in clinical machine learning*. *GigaScience*, 6(5), gix019.
<https://doi.org/10.1093/gigascience/gix019>